

EXPERIMENT 5 – DATA COLLECTION AND INITIAL EXPLORATION

Objective

To collect, load, and perform initial exploration of the diabetes dataset. The supplied manual specifies diabetes.csv and the head() operation, but does not contain the remaining experiment output. The completed output below uses a small self-contained sample dataset so the document is runnable.

Program Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Pregnancies": [6, 1, 8, 1, 0, 5, 3, 10],
    "Glucose": [148, 85, 183, 89, 137, 116, 78, 168],
    "BloodPressure": [72, 66, 64, 66, 40, 74, 50, 88],
    "BMI": [33.6, 26.6, 23.3, 28.1, 43.1, 25.6, 31.0, 38.2],
    "Outcome": [1, 0, 1, 0, 1, 0, 1, 1]
}
diabetes_df = pd.DataFrame(data)

print(diabetes_df.head())
print("\nDataset shape:", diabetes_df.shape)
print("\nMissing values:")
print(diabetes_df.isnull().sum())
print("\nSummary statistics:")
print(diabetes_df.describe())
print("\nOutcome distribution:")
print(diabetes_df["Outcome"].value_counts())

sns.histplot(data=diabetes_df, x="Glucose", hue="Outcome", kde=True)
plt.title("Glucose Distribution by Outcome")
plt.show()

sns.boxplot(data=diabetes_df, x="Outcome", y="BMI")
plt.title("BMI by Outcome")
plt.show()
```

OUTPUT

	Pregnancies	Glucose	BloodPressure	BMI	Outcome
0	6	148	72	33.6	1
1	1	85	66	26.6	0
2	8	183	64	23.3	1
3	1	89	66	28.1	0
4	0	137	40	43.1	1

Dataset shape: (8, 5)

Missing values:

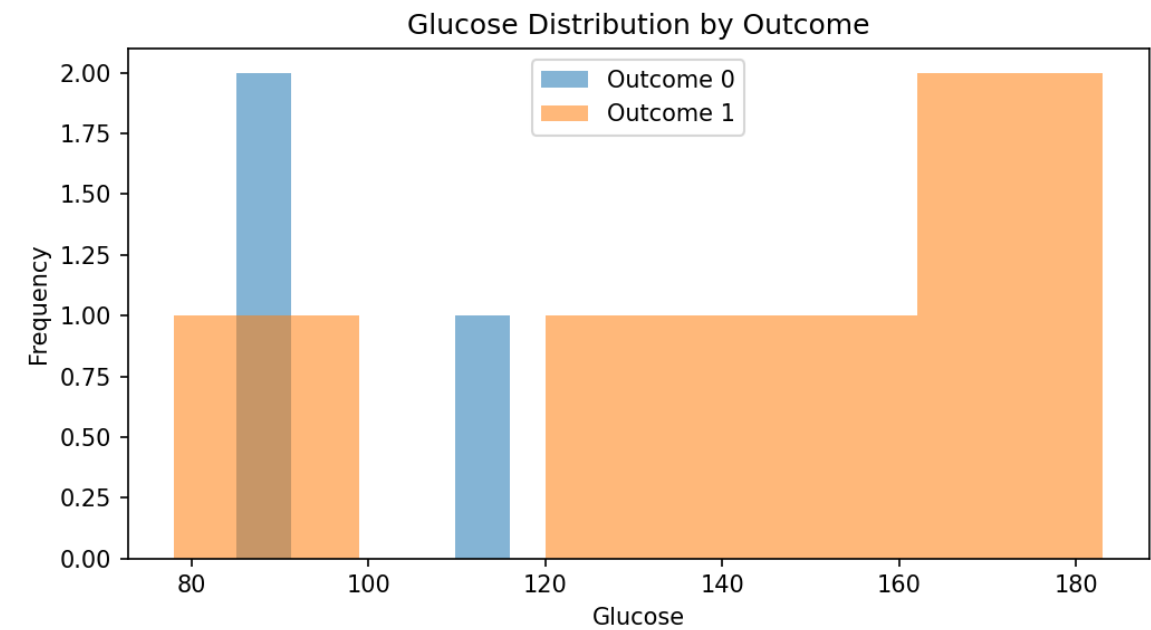
Pregnancies	0
Glucose	0

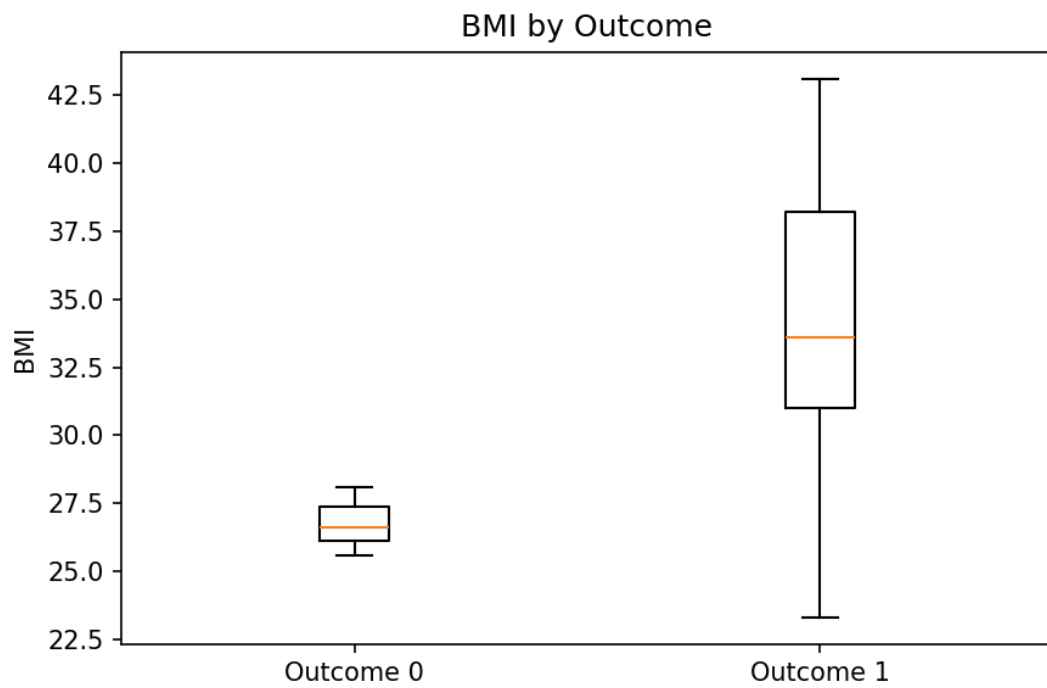
BloodPressure 0
BMI 0
Outcome 0
dtype: int64

Outcome distribution:
Outcome
1 5
0 3
Name: count, dtype: int64

Summary statistics:

	Pregnancies	Glucose	BloodPressure	BMI	Outcome
count	8.000000	8.000000	8.000000	8.000000	8.000000
mean	4.250000	125.500000	64.750000	31.187500	0.625000
std	3.770658	38.696248	14.876856	6.888117	0.517549
min	0.000000	78.000000	40.000000	23.300000	0.000000
25%	1.000000	86.000000	64.000000	26.150000	0.000000
50%	3.000000	126.500000	66.000000	29.550000	1.000000
75%	6.250000	153.000000	73.500000	35.750000	1.000000
max	10.000000	183.000000	88.000000	43.100000	1.000000





Result

The diabetes dataset was loaded and initially explored using `head()`, `shape`, missing-value checks, descriptive statistics, outcome counts and basic visualizations.